



Solving linear equations of the form $ax + b = cx + d$

Teacher resource

Question 1

What value of m makes this equation true?

$$2m + 17 = 5m - 4$$

Answer: 7

Skill: Students solve linear equations.

Background information

Some students may not identify this form of a linear equation and will require explicit support in the use of the inverse relationship between addition and subtraction and the inverse relationship between multiplication and division to rearrange and simplify the equation.

Additional questions

1. What value of m makes the equation $5m + 17 = 2m - 4$ true?
2. What value of m makes the equation $5m - 4 = 2m + 17$ true?

Question 2

$$4(3c - 6) = 2c + 11$$

What is the value of c in this equation?

Answer: 3.5

Skill: Students expand brackets to solve a linear equation.

Additional questions

1. What value of c makes the equation $2(3c - 6) = 4c + 11$ true?
2. If $c = 5$, what symbol ($<$, $>$, $=$) makes the statement $2(3c - 6) \square 4c + 11$ true?

Question 3

Alex and Pat both decide to save money.
Alex begins with \$200 and saves \$40 each week.
Pat begins with \$500 and saves \$10 each week.

After how many weeks will Alex and Pat have equal amounts of money?

Answer: 10

Skill: Students model a given situation with a linear equation

Additional questions

1. How much money will they each have at this time?
2. After how many weeks will Alex have twice as much as Pat?

Question 4

The length of a side of a square is x cm. The length of one side of an equilateral triangle is $(3x - 5)$ cm.

The perimeter of the square is equal to the perimeter of the equilateral triangle.

What is the area of the square?

Answer: 9 cm^2

Skill: Students construct and solve a linear equation.

Background information:

This question is an example of a two-step problem. Students may need support to understand that the information needed to calculate the area of the square, the length of a side of the square, is not stated in the question. As the first step in solving this problem, students need to use the given information to find the length of the side of the square.

Additional questions

1. What is the perimeter of the equilateral triangle?
2. Let $ABCD$ be a rectangle with side CD three times as long as side AD .
If the length of CD is $(5x - 7)$ cm and the length of AD is $(x + 1)$ cm, what is its area?

Student activity: Solving linear equations of the form $ax + b = cx + d$

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